

## Laboratory and glasshouse evaluation of entomopathogenic fungi against the two-spotted spider mite, *Tetranychus urticae* (Acari: Tetranychidae), on tomato, *Lycopersicon esculentum*

D. CHANDLER<sup>1</sup>, G. DAVIDSON<sup>1</sup>, & R. J. JACOBSON<sup>2</sup>

<sup>1</sup>Warwick HRI, Wellesbourne, Warwick, UK, and <sup>2</sup>Stockbridge Technology Centre, Caxwood, N. Yorkshire, UK

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### Abstract

Forty isolates of entomopathogenic fungi from six genera were assessed against the two-spotted spider mite, *Tetranychus urticae*, in a single dose, direct application laboratory bioassay on tomato leaflets. Only three isolates caused greater mortality than the control: these were *Metarhizium anisopliae* 442.99, *Hirsutella* spp. 457.99, and *Verticillium lecanii* 450.99. These isolates were assessed in a multiple dose bioassay, together with three isolates cultured from commercial biopesticides as follows: *Beauveria bassiana* 432.99 (cultured from 'Naturalis-L', Troy Biosciences, Phoenix, TX, USA); *Hirsutella thompsonii* 463.99 (cultured from 'Mycar', Abbott Laboratories USA); and *V. lecanii* 19.79 (used in 'Mycotal' Koppert BV, The Netherlands). *Beauveria bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, and *V. lecanii* 450.99 were all pathogenic to *T. urticae* in this bioassay. In addition, it was found that the mortality caused by *B. bassiana* 432.99 and Naturalis-L was increased when the mites were exposed to tomato leaflets sprayed previously with conidia suspensions, compared to spraying the mites directly. In a glasshouse experiment, sprays of *B. bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, *V. lecanii* 450.99 and Naturalis-L reduced *T. urticae* populations in a tomato crop grown according to commercial practice. Naturalis-L reduced *T. urticae* numbers by up to 97%. In a second glasshouse experiment, single sprays of Naturalis-L and the acaricide fenbutatin oxide (Torq) were compared as supplementary treatments to release of the predatory mite, *Phytoseiulus persimilis*. Supplementary sprays of fenbutatin oxide reduced the numbers of *T. urticae* nymphs (80% reduction), while Naturalis-L reduced numbers of *T. urticae* adults, nymphs and eggs (98% reduction in all three cases). It is concluded that Naturalis-L has the potential to be used against *T. urticae* on glasshouse tomato crops.

**Keywords:** Two-spotted spider mite, *Tetranychus urticae*, tomato, *Lycopersicon esculentum*, entomopathogenic fungi, *Metarhizium anisopliae*, *Beauveria bassiana*, *Verticillium lecanii*, *Hirsutella*

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Correspondence: D. Chandler, Warwick HRI, Wellesbourne, Warwick, CV35 9EF, UK. Tel: +44-24-7657-4455. Fax: +44-24-7657-4500. E-mail: dave.chandler@warwick.ac.uk

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## Introduction

The two-spotted spider mite, *Tetranychus urticae* Koch (Acari, Tetranychidae) is a major pest of crops world-wide (Walter & Proctor, 1999). It is polyphagous and has evolved resistance to many chemical pesticides (Cranham & Helle, 1985). In response to pesticide resistance, farmers and growers have increased their use of biological control, which is done by conserving natural enemies and/or by applying predatory phytoseiid mites. However, this is often not sufficient, and supplementary sprays of acaricides are needed on a routine basis, although this is unlikely to be sustainable given the propensity of *T. urticae* to develop resistance. There is a need for an effective method of *T. urticae* control that does not involve chemicals, to circumvent the problems of acaricide resistance and also to enable farmers and growers to respond to consumer concerns about pesticide residues. This is most likely to be achieved with a suite of natural enemies that complement one another's activities at different times during crop and pest development (Jacobson, 1999). In particular, it should include a fast-acting microbial biopesticide to replace the chemical acaricides that are used currently as supplementary treatments. Such use of a microbial biopesticide has worked well for control of other pests, for example western flower thrips, *Frankliniella occidentalis*, on cucumbers using entomopathogenic fungi and the predatory mite *Neoseiulus* (*Amblyseius*) *cucumeris* (Jacobson et al., 2001).

The most promising microbial control agents of *T. urticae* are entomopathogenic fungi, which invade their hosts by growing through the cuticle. They are able, therefore, to infect sap-feeding pests, such as *T. urticae*, which are unlikely to acquire a pathogen *per os*. In this paper, we report on a series of laboratory and glasshouse experiments to investigate the susceptibility of *T. urticae* to anamorphic entomopathogenic fungi on glasshouse-grown tomato. *Tetranychus urticae* is a particular problem on this crop in the UK, where control is based on applications of the predatory mite, *Phytoseiulus persimilis*, reinforced with sprays of a selective acaricide, fenbutatin oxide, although resistance has started to develop (Jacobson et al., 1999).

## Materials and methods

### *Fungal isolates*

Forty isolates of fungi from six genera were used in the study (Table I): *Beauveria*, *Hirsutella*, *Metarhizium*, *Paecilomyces*, *Tolypocladium* and *Verticillium*. The isolates are referred to by the accession number used in the culture collection at Warwick HRI. Most of the isolates originated from mites or ticks while others originated from insect hosts, but were reported from the literature or personal communications to kill mites. Isolates cultured from eight commercial biopesticides were also included. Stock cultures of the isolates were stored in liquid nitrogen vapour (Chandler, 1994) in the Warwick HRI culture collection. Laboratory cultures were grown on Sabouraud's dextrose agar (SDA) slopes and maintained in a refrigerator at 4°C for up to 6 months. For bioassays, subcultures were prepared on SDA from the refrigerated slope cultures and incubated at either  $26 \pm 1^\circ\text{C}$  for 18–21 days (*Hirsutella* isolates) or  $23 \pm 1^\circ\text{C}$  for 10–14 days (all other isolates) in the dark. Conidia were harvested from the subcultures in sterile 0.01% Triton X-100 and filtered through sintered glass thimbles (40–100 µm pore). Conidia were enumerated using an improved Neubauer haemocytometer and aliquots (10 mL) were prepared at concentrations of  $3 \times 10^6$

Table I. Isolates used in the study. All isolates were examined in Laboratory Experiment 1 and selected isolates were evaluated in Laboratory Experiments 2 and 3

| Species                          | Isolate number <sup>†</sup>      | Host or source                                             | Country    |
|----------------------------------|----------------------------------|------------------------------------------------------------|------------|
| <i>Beauveria bassiana</i>        | 431.99 (T228) <sup>a</sup>       | Acari                                                      | Denmark    |
|                                  | 432.99 (JW-1) <sup>b</sup>       | <i>Anthonomus grandis</i> (Coleoptera: Curculionidae)      | USA        |
|                                  | 433.99 <sup>c</sup>              | –                                                          | –          |
|                                  | 434.99 (ARSEF2869) <sup>d</sup>  | <i>Bephratelloides cubensis</i> (Hymenoptera: Eurytomidae) | USA        |
|                                  | 454.99 (I91 636) <sup>e</sup>    | Acari                                                      | –          |
|                                  | 455.99 (DAT 049) <sup>f</sup>    | –                                                          | –          |
|                                  | 460.99 <sup>h</sup>              | Acari                                                      | Israel     |
| <i>Hirsutella</i> sp.            | 436.99 (H2) <sup>i</sup>         | Acari: Tarsonemidae                                        | Poland     |
|                                  | 437.99 (H3) <sup>i</sup>         | <i>Eriophyes piri</i> (Acari: Eriophyidae)                 | Poland     |
|                                  | 438.99 (H4) <sup>i</sup>         | <i>Abacarus hystrix</i> (Acari: Eriophyidae)               | Poland     |
|                                  | 440.99 (H12) <sup>i</sup>        | <i>Stenotarsonemus fragariae</i> (Acari: Tarsonemidae)     | Poland     |
|                                  | 457.99 (3339C) <sup>j</sup>      | <i>Dendrolaelaps</i> spp. (Acari: Digamasellidae)          | Poland     |
| <i>Hirsutella necatrix</i>       | 49.81 (IMI252317) <sup>e</sup>   | <i>Abacarus hystrix</i> (Acari: Eriophyidae)               | UK         |
| <i>Hirsutella thompsonii</i>     | 34.79                            | <i>Eriophyes guerreronis</i> (Acari: Eriophyidae)          | USA        |
|                                  | 51.81 <sup>k</sup>               | <i>Eriophyes guerreronis</i> (Acari: Eriophyidae)          | USA        |
|                                  | 71.82                            | <i>Eriophyes guerreronis</i> (Acari: Eriophyidae)          | Jamaica    |
|                                  | 73.82 <sup>k</sup>               | <i>Phyllocoptura captrila</i> (Acari: Eriophyidae)         | USA        |
|                                  | 75.82 <sup>k</sup>               | <i>Colomerus novaehbridensis</i> (Acari: Eriophyidae)      | New Guinea |
|                                  | 77.82 <sup>k</sup>               | <i>Eriophyes guerreronis</i> (Acari: Eriophyidae)          | Jamaica    |
|                                  | 463.99 (DoCo AL) <sup>l,m</sup>  | <i>Varroa jacobsonii</i> (Acari: Varroidae)                | Canada     |
| <i>Metarhizium anisopliae</i>    | 275.86 <sup>n</sup>              | <i>Cydia pomonella</i> (Lepidoptera: Tortricidae)          | Germany    |
|                                  | 392.93                           | Soil                                                       | UK         |
|                                  | 441.99 (ARSEF3297) <sup>d</sup>  | <i>Boophilus</i> spp. (Acari: Ixodidae)                    | Mexico     |
|                                  | 442.99 (ARSEF4556) <sup>d</sup>  | <i>Boophilus</i> spp. (Acari: Ixodidae)                    | USA        |
|                                  | 444.99 (ATCC38249) <sup>o</sup>  | <i>Hylobius pales</i> (Coleoptera: Curculionidae)          | –          |
|                                  | 445.99 <sup>p</sup>              | –                                                          | –          |
|                                  | 456.99 (DAT F001) <sup>f,g</sup> | <i>Adoryphorus couloni</i> (Coleoptera: Scarabaeidae)      | Australia  |
| <i>Paecilomyces farinosus</i>    | 446.99 (CCFC002085) <sup>q</sup> | <i>Mycobates</i> spp. (Acari: Mycobatidae)                 | Canada     |
| <i>Paecilomyces fumosoroseus</i> | 409.96 <sup>r</sup>              | <i>Phenacoccus solani</i> (Hemiptera: Pseudococcidae)      | USA        |
|                                  | 447.99 (KVL319) <sup>a</sup>     | <i>Ixodes ricinus</i> (Acari: Ixodidae)                    | –          |
| <i>Tolypocladium inflatum</i>    | 448.99 (ARSEF3278) <sup>d</sup>  | <i>Mycobates</i> spp. (Acari: Mycobatidae)                 | Canada     |
| <i>Tolypocladium niveum</i>      | 449.99 (CCFC002081) <sup>q</sup> | <i>Mycobates</i> spp. (Acari: Mycobatidae)                 | Canada     |
| <i>Verticillium lecanii</i>      | 1.72 <sup>s</sup>                | <i>Macrosiphoniella sanborni</i> (Homoptera: Aphididae)    | UK         |
|                                  | 17.76                            | <i>Cecidophyopsis</i> spp. (Acari: Eriophyidae)            | UK         |
|                                  | 19.79 <sup>t</sup>               | <i>Trialeurodes vaporariorum</i> (Homoptera: Aleyrodidae)  | UK         |
|                                  | 30.79                            | <i>Cecidophyopsis</i> spp. (Acari: Eriophyidae)            | UK         |
|                                  | 31.79                            | <i>Cecidophyopsis</i> spp. (Acari: Eriophyidae)            | UK         |
|                                  | 450.99 (IMI235048) <sup>e</sup>  | <i>Cecidophyopsis ribis</i> (Acari: Eriophyidae)           | UK         |
|                                  | 452.99 (CBS317.70A) <sup>u</sup> | <i>Tetranychus urticae</i> (Acari: Tetranychidae)          | –          |
|                                  | 453.99 (CCFC006079) <sup>q</sup> | Acari                                                      | Canada     |

<sup>†</sup>Isolate number in the Warwick HRI culture collection (isolate number from culture collection of origin).

<sup>a</sup>Kindly supplied by T. Steenberg, Danish Pest Infestation Laboratory, Skovbrynet 14, DK-2800, Lyngby, Denmark.

<sup>b</sup>Isolate found in the commercial product Naturalis-L (Troy Biosciences Inc., 113 South 47th Avenue, Phoenix, AZ 850433, USA).

<sup>c</sup>Isolate found in the commercial product BotaniGard (Mycotech Corporation, P.O. Box 4109, Butte, MT 59702, USA).

<sup>d</sup>Kindly supplied by the USDA-ARS Collection of Entomopathogenic Fungal Cultures (ARSEF), USDA-ARS Plant Protection Research Unit, US Plant, Soil and Nutrition Laboratory, Tower Road, Ithaca, NY 14853-2901, USA.

<sup>e</sup>Obtained from the CAB International Mycological Institute, Bakeham Lane, Egham, Surrey, UK.

<sup>f</sup>Kindly supplied by Bio-Care Technology Pty Ltd., RMB 1084 Pacific Highway, Somersby, NSW 2250, Australia.

<sup>g</sup>Isolate found in the commercial product Biogreen (Bio-Care Technology Pty. Ltd., R.M.B. 1084 Pacific Highway, Somersby, NSW 2250, Australia).

<sup>h</sup>Kindly supplied by M. Samish, Division of Parasitology, Kimron Veterinary Institute, Beit Dagan 50250, P.O. Box 12, Israel.

<sup>i</sup>Kindly supplied by C. Tkaczuk, University of Podlasie, Department of Plant Protection, ul. Prusa 14, 08110 Siedlce, Poland.

<sup>j</sup>Kindly supplied by S. Balazy, Research Centre for Agricultural and Forest Environment of the Polish Academy of Sciences, ul. Bukowska 19, 60-809 Poznan, Poland.

<sup>k</sup>Kindly supplied by C.W. McCoy, University of Florida, CREC, 700 Experiment Station Road, Lake Alfred, FL 33850, USA.

Table 1 (continued)

<sup>l</sup>Isolate found in the former commercial product Mycar (Abbott Laboratories, 100 Abbott Park, Road, Abbott Park, IL 60063-3500, USA).

<sup>m</sup>Kindly supplied by B. Ruzicka, 2910 Glenmore Road North, Kelowna, British Columbia, Canada V1V 2B6.

<sup>n</sup>Kindly supplied by G. Zimmermann, Biologische Bundesanstalt f. Land-und Forstwirtschaft, Institut f. Biologischen Pflanzenschutz, Heinrichstr. 243, D-64287 Darmstadt, Germany.

<sup>o</sup>Obtained from the American Type Culture Collection, 10801 University Boulevard, Manassas, VA 20110-2209, USA.

<sup>p</sup>Isolate found in the commercial product Bio-Blast (Eco-Science Corporation, 17 Christopher Way, Eatontown, NJ 07724, USA).

<sup>q</sup>Obtained from the Canadian Collection of Fungal Cultures, ECORC Room 1015, K.W. Neatby building, C.E.F. Ottawa, Ontario, Canada.

<sup>r</sup>Isolate found in the commercial product PFR'97 (Thermo-Trilogy Corporation, 9145 Guildford Road, Suite 175, Columbia, MD 21046, USA).

<sup>s</sup>Isolate found in the commercial product Vertalec (Koppert Biological Systems, P.O. Box 155, 2650 AD Berkel en Rodenrijs, The Netherlands).

<sup>t</sup>Isolate found in the commercial product Mycotol (Koppert Biological Systems, P.O. Box 155, 2650 AD Berkel en Rodenrijs, The Netherlands).

<sup>u</sup>Obtained from the Centraalbureau voor Schimmelcultures, P.O. Box 273, 3740AG Baarn, The Netherlands.

to  $3 \times 10^8$  conidia  $\text{mL}^{-1}$  depending on the experiment. Conidia germination was measured on SDA after incubation for 24 h at 26/23°C (Goettel & Inglis, 1997). Isolates from the genera *Beauveria*, *Metarhizium*, *Paecilomyces* and *Verticillium* exhibited an average germination of >90%, with the exception of isolate *M. anisopliae* 445.99 (86% germination). The average germination of the two isolates of *Tolypocladium* were 60% (*Tolypocladium inflatum* 448.99) and 65% (*Tolypocladium niveum* 449.99). The average germination of eight out of the 13 *Hirsutella* isolates was <90% as follows: (1) *Hirsutella* sp. 436.99 (84%); (2) *Hirsutella* sp. 437.99 (89%); (3) *Hirsutella* sp. 438.99 (71%); (4) *Hirsutella* sp. 440.99 (70%); (5) *Hirsutella* sp. 457.99 (37%); (6) *Hirsutella necatrix* 49.81 (67%); (7) *Hirsutella thompsonii* 34.79 (74%); and (8). *H. thompsonii* 71.82 (83%).

The commercial biopesticide Naturalis-L (Troy Biosciences Inc., Phoenix, USA), which contains the JW-1 isolate of *Beauveria bassiana* as its active ingredient (ARSEF 3097 = ATCC 7404 = FCI 7744 = Warwick HRI isolate 432.99), was used in later experiments. The concentration of conidia in Naturalis-L =  $2.5 \times 10^8$   $\text{mL}^{-1}$  product. Naturalis-L was stored at 5°C before use and prepared according to the manufacturer's instructions.

### Tetranychus urticae cultures

Cultures of *T. urticae* were reared on tomato, *Lycopersicon esculentum* cv Spectra. This cv was used also in laboratory bioassays. Tomato seeds were germinated for 14 days in rockwool plugs (Avoncrop Ltd., Bristol, UK) ( $4 \times 4 \times 4$  cm) containing vermiculite in a glasshouse at 25°C and at least 16 h of light per day. The seedlings were then potted up in Levington F2 compost (Scotts UK, Ipswich, UK) and grown in the glasshouse for approximately 5 weeks until 30–50 cm tall.

Stock cultures of *T. urticae* were reared in a glasshouse (20–30°C, 50–70% RH, and at least 16 h of light per day). New colonies were initiated weekly by placing a *T. urticae*-colonised tomato leaf onto a leaf of a new culture plant. The laboratory bioassays were done with single age populations of *T. urticae*. Batches of 15 adult female *T. urticae* were collected by hand from stock culture plants using a dissecting needle, then placed on inverted tomato leaflets on damp filter paper in Petri dishes (90-mm diameter, triple vented), and incubated at 21°C, L:D 16:8 h for 48 h to allow

oviposition (Vindon incubator cabinet, Vindon Scientific Ltd., Oldham, UK). The adult mites were removed after this period using a dissecting needle and the leaflets were incubated for a further 6 days. Leaflets with spider mite nymphs were then placed on whole tomato plants in the glasshouse for 7–9 days. The plants were monitored until new adults were found. Adult females that had just begun to produce eggs were then selected for experiments.

### *Laboratory experiments*

*Laboratory Experiment 1: susceptibility of T. urticae to 40 isolates of entomopathogenic fungi.* Forty isolates of fungi (Table I) were screened against *T. urticae* in a replicated, single-concentration bioassay (concentration =  $1 \times 10^7$  conidia mL<sup>-1</sup>). Groups of 25–30 single age female *T. urticae* were collected from tomato plants by hand, using a dissecting needle. The mites were placed on a roughened glass cover slip (22 × 22 mm) on water-saturated filter paper within the base of a Petri dish (90-mm diameter, triple vented). Saturating the filter paper with water helped to confine the mites to the cover slip. The cover slips were roughened by rubbing them with a kitchen scouring pad. This provided a surface on which the mites could grip and prevented them being dislodged during spraying. The mites were sprayed with 2 mL of a suspension of fungal conidia, using a Potter tower (Potter, 1952) with an ‘intermediate’ atomiser and a spray pressure of 50 kPa. Controls consisted of 0.01% Triton X-100. Each cover slip was then placed on an inverted tomato leaflet (cv Spectra) on dampened filter paper within a ventilated Petri dish (90-mm diameter, two mesh-covered ventilation panels (0.17 × 0.37 mm per dish) and air dried within a laminar flow cabinet for 10 min. Most of the mites vacated the cover slip onto the leaflet during this time, but any remaining were transferred by tapping the cover slip gently with a dissecting needle. The number of mites placed on each leaflet was recorded, and the Petri dishes were sealed with Nescofilm. The Petri dishes were placed within an airtight perspex box (Fisher Scientific, Loughborough, UK, 30.5 × 30.5 × 45.7 cm) containing 750 mL distilled water in a tray at the bottom to maintain a saturated atmosphere. The box was maintained in an incubator (Vindon incubator cabinet, Vindon Scientific Ltd., Oldham, UK) at 23°C, L:D 16:8 h. Ventilating the Petri dishes prevented the accumulation of droplets of water on the leaflets in which the mites could become trapped. The numbers of living and dead mites (no movement when touched with a dissecting needle) were counted daily for 9 days. Cadavers were removed onto damp filter paper within Petri dishes sealed with Nescofilm and examined for the appearance of sporulating mycelium on the integument 7 days after the end of the bioassay.

The 40 fungal isolates were bioassayed according to an alpha design (Patterson & Williams, 1976) done as four blocks of 10 isolates, replicated three times. Each block of each replicate was done on a separate occasion, so that bioassays were done on 12 occasions in total. Each fungal treatment consisted of one Petri dish of *T. urticae* per replicate. Each block included four control dishes, except for the first two occasions where two control dishes were used. Percentage mortalities at 6 days post inoculation were transformed to logits and analysed using the residual maximum likelihood (REML) algorithm (Patterson & Thompson, 1971) in Genstat (2000). Control mortalities from different blocks were also analysed by ANOVA (Genstat, 2000).

*Laboratory Experiment 2: susceptibility of T. urticae to six fungal isolates in a multiple dose bioassay.* Six isolates were selected for assessment in a multiple dose laboratory bioassay, including three isolates from commercial biopesticides: *B. bassiana* 432.99 (Naturalis-L, Troy Biosciences Inc.); *Hirsutella* sp. 457.99; *H. thompsonii* 463.99 (Mycar, Abbott Laboratories, IL, USA); *Metarhizium anisopliae* 442.99; *Verticillium lecanii* 19.79 (Mycotal, Koppert BV, Berkel en Rodenrijs, The Netherlands); and *V. lecanii* 450.99. The bioassay procedure was as described in Experiment 1 and used 22–27 mites per batch. Five concentrations of conidia were used, from  $3 \times 10^6$  to  $3 \times 10^8$  mL<sup>-1</sup>. An extra cover slip was placed next to the mites during spraying in order to estimate doses of conidia. This was done by washing conidia from each cover slip in 10 mL 0.01% Triton X-100, plating 0.1 mL aliquots of the resulting suspension onto SDA after serial dilution, and counting numbers of colony forming units (cfu) following incubation at 26/23°C, after the method of Goettel & Inglis (1997).

The six fungal isolates were bioassayed according to a resolvable incomplete block design, with three blocks of two isolates, replicated three times. Each block of each replicate was done on a separate occasion, so that bioassays were done on nine occasions in total. Each isolate was bioassayed as one Petri dish of *T. urticae* per concentration per block. Each block included four control bioassay dishes. Dose response was analysed at 6 days post inoculation. Control mortality in each block was calculated with a minimum of 88 mites, hence we felt there was sufficient information to adjust the data for control mortality using Abbott's formula (Abbott, 1925), according to  $T = tc$  and  $D = \max\{T - a, 0\}$ , where  $T$  is the adjusted total number of mites on a treated leaflet,  $t$  is the total number of mites per leaflet,  $c$  is the proportion of mites alive on control leaflets at 6 days post inoculation,  $D$  is the adjusted total of dead mites on a treated leaflet, and  $a$  is the number of mites alive on a treated leaflet at 6 days post inoculation. A generalised linear model with a logit link function was used to describe the relationship between  $D$  and log<sub>10</sub> conidia concentration for each isolate separately (Genstat, 2000). Mortalities were low overall, and although the median lethal concentration (LC<sub>50</sub>) could be calculated for four of the six isolates, only one was within the experimental range, and hence the concentration for 25% mortality (LC<sub>25</sub>) was estimated instead. The LC<sub>25</sub> and its fiducial limits were estimated using Fieller's Theorem (Finney, 1971).

*Laboratory Experiment 3: comparison of direct application and indirect application bioassays.* The direct application bioassay used in Laboratory Experiments 1 and 2 was compared with an indirect application bioassay. Suspensions of conidia (concentration =  $1 \times 10^8$  mL<sup>-1</sup>) were prepared for four isolates of fungi: *B. bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, and *V. lecanii* 450.99. Naturalis-L (active ingredient *B. bassiana* JW-1/Warwick HRI isolate 432.99) was included as an additional treatment, and was applied at the manufacturer's recommended concentration ( $1 \times 10^5$  mL<sup>-1</sup>). For the direct application method, groups of 19–25 single-age female *T. urticae* were sprayed with 2 mL of a suspension of conidia using a Potter tower, as described previously. For the indirect application method, tomato leaflets were sprayed with 2 mL of suspension of fungal conidia using a Potter tower at 50 kPa (controls treated with 0.01% Triton X-100) and placed on damp filter paper within a Petri dish (90-mm diameter, triple vented). Groups of 19–25 single-age adult female *T. urticae*, which had been collected previously and placed onto glass cover slips on water saturated filter paper within the base of a Petri dish, were then transferred onto

the lower surface of each leaflet by tapping the cover slip gently. The dishes were left to dry on the laboratory bench for 1 h, then sealed with Nescofilm. The dishes were maintained in an incubator at 23°C, L:D 16:8 h for the duration of the bioassay. Mortality was assessed daily for 7 days. All fungal isolates were bioassayed concurrently as a block and this was replicated on six separate occasions. On each occasion, two control dishes per application method were included. Data were adjusted to take account of control mortality as described for Experiment 2 and percentage mortalities calculated at 6 days post inoculation. Data were compared by ANOVA incorporating an angular transformation (Genstat, 2000).

### *Glasshouse experiments*

*Crop raising.* Glasshouse experiments were done in long season tomato crops (*L. esculentum* cv Espero) grown hydroponically in rockwool slabs. The crops were grown in the compartments of a 1456 m<sup>2</sup> glasshouse. Each compartment measured 10 × 9.3 × 4.2 m. The compartments were maintained at a minimum temperature of 17°C and vented at temperatures ranging from 19 to 26°C depending on the age of the crop. Glasshouse Experiment 1 used one compartment, and Glasshouse Experiment 2 used two adjacent compartments. The floor of each compartment was covered with white plastic sheeting to reflect light and minimise the risk from soil borne plant pathogens. The crop in each compartment consisted of six double rows of plants. Each double row contained 24 plants grown in six, 120-cm rockwool slabs (i.e., four plants per slab) and trained in a V system, which therefore formed the double row. Irrigation with nutrient solution, CO<sub>2</sub> dosing, and plant husbandry (training, tying in, deleafing, layering, and fruit picking) were done according to standard commercial practice (Van de Vooren et al., 1986; Adams & Valdés, 2002). Pollination was done with bumblebees (BCP Ltd., Ashford, Kent, UK). No additional chemical insecticides or acaricides were applied during the experiments, although a fungicide treatment of Thiovit (Novartis, Cambridge, UK) was applied in Experiment 2 on 13 May 2002 to control powdery mildew.

*Phytoseiulus persimilis* cultures. *Phytoseiulus persimilis* cultures were obtained from BCP Ltd. The cultures were supplied in 30-mL dispenser tubes containing ca. 2000 individuals in approximately 25 mL of vermiculite carrier. The cultures were stored at 10°C for a maximum of 24 h prior to use. *Phytoseiulus persimilis* were dispensed onto tomato leaves by tapping the tube once over each leaf. The number of mites dispensed in this way was estimated to be four to six mites per leaf. This estimate was in keeping with that stated by the manufacturer. The manufacturer's recommended release rate was five to 20 *P. persimilis* per m<sup>2</sup> depending on the level of *T. urticae* infestation.

*Fenbutatin oxide.* The selective acaricide fenbutatin oxide was obtained as the commercial product Torq (Fargro Ltd., Littlehampton, UK). It was stored according to the manufacturer's recommendations and was applied at the manufacturer's recommended concentration (0.5 g L<sup>-1</sup>).

*Glasshouse Experiment 1: efficacy of entomopathogenic fungi against T. urticae populations on tomato.* Glasshouse Experiment 1 was planted on 16 March 2001. The experiment comprised six treatments: (1) untreated control; (2) *B. bassiana* 432.99; (3) *H.*

*thompsonii* 463.99; (4) *M. anisopliae* 442.99; (5) *V. lecanii* 450.99; and (6) Naturalis-L ( $\equiv B. bassiana$  432.99). Suspensions of fungal conidia ( $1 \times 10^8 \text{ mL}^{-1}$ ) were prepared as described previously from all isolates except Naturalis-L, which was prepared at the manufacturer's recommended rate ( $1 \times 10^5 \text{ conidia mL}^{-1}$ , equivalent to  $0.5 \text{ g L}^{-1}$  water).

The experiment was done according to a randomised block design. The treatments were applied to tomato leaves on one side of a double row of plants. This was done in a total of four rows in the glasshouse. Each of these rows received all six of the treatments as a block. The rows receiving the treatments were separated from each other by guard rows. There was one treatment per plant. Each treatment was applied to three marked leaves per plant, at the top (3.5 m from the ground), middle (2.5 m) and bottom (1.5 m) of the plant. There was a total of 12 leaves per treatment in the glasshouse. Each treated plant was separated by two guard plants.

The experiment was started on 4 September 2001. Adult female *T. urticae* were released onto one leaflet of each marked leaf, 40 per leaf, using a pooter with a detachable tip that could be left on the leaf, secured using a small amount of Vaseline on the leaf surface. This was followed by a further release of 20 mites per leaf, 7 days later. Fungal treatments were applied 7 days after the second release of *T. urticae*. Suspensions of conidia were sprayed to run-off onto the leaflet on which the mites were released. Sprays were applied with a 500 mL hand held sprayer (Cherwell, Southam, UK). The floor of the house was hosed lightly with water to increase humidity and simulate a whole crop spray. Spraying was done in the late afternoon. Sprays of conidia suspensions were applied twice, with a 7-day interval. Seven days after the second spray, leaves were removed and the numbers of motile *T. urticae* (nymphs+adults) and eggs per leaf were recorded. The transformation  $\log_{10}(\text{total} + 0.375)$  was applied to the total numbers of motile *T. urticae* and numbers of eggs before ANOVA (Genstat, 2000). The volume of spray deposited on leaves was estimated at  $7.8 \mu\text{L cm}^{-2}$  (SD = 5.15), obtained by measuring the surface area of 20 tomato leaves using a Delta-T Leaf Area Meter Mk 2 (Delta-T Devices Ltd., Cambridge, UK), spraying the leaves to run-off with 0.01% Triton X-100 using the hand held sprayer, and weighing the amount of liquid applied per leaf.

*Glasshouse Experiment 2: efficacy of Naturalis-L as a supplementary treatment to the predatory mite, P. persimilis, for control of T. urticae populations on tomato.* Glasshouse Experiment 2 was planted on 20 March 2002. This experiment was done to examine the effectiveness of Naturalis-L as a supplementary treatment to *P. persimilis*, in comparison with the selective acaricide, fenbutatin oxide. There were four treatments: (1) untreated control; (2) *P. persimilis*; (3) *P. persimilis* + fenbutatin oxide; and (4) *P. persimilis* + Naturalis-L. Fenbutatin oxide and Naturalis-L were prepared at the manufacturer's recommended rates ( $0.5 \text{ g L}^{-1}$  and  $1 \times 10^5 \text{ conidia mL}^{-1}$ , respectively). Low numbers of *P. persimilis* were introduced into the crop to simulate a situation in which control with *P. persimilis* was failing, and hence sprays of fenbutatin oxide/Naturalis-L were needed as a supplementary treatment.

The experiment was done according to a randomised block design within two adjacent glasshouse compartments. The treatments were applied to tomato leaves on one side of a double row of plants, in a total of four rows in each glasshouse. Each of these rows contained two of the four treatments. The treated rows were separated



from each other by guard rows. There was a block of six plants for each treatment in a row. Each block was separated by two guard plants. Each treatment was applied to one marked leaf near the top (3 m from the ground) of each of the six plants. There was a total of 24 leaves per treatment.

The experiment was done on the 5 July 2002. Adult female *T. urticae* were released onto one leaflet of each marked leaf, 25 per leaf, using *T. urticae*-colonised leaf material from the stock culture. This was followed by a further release of *T. urticae* 14 days later. *Phytoseiulus persimilis* were introduced 7 days after the second release of *T. urticae*, at a rate of four to six per leaf, as described previously. Fenbutatin oxide and Naturalis-L were sprayed to run-off onto marked leaves, 4 days after the release of *P. persimilis*, and the floor of the house was hosed lightly with water, as described in Glasshouse Experiment 1. Seven days after the application of fenbutatin oxide and Naturalis-L, leaves were removed and numbers of *T. urticae* eggs, nymphs and adults, and numbers of *P. persimilis*, were recorded. The transformation  $\log_{10}(\text{total} + 0.375)$  was applied to the total numbers of *T. urticae* eggs, nymphs and adults, and the total numbers of *P. persimilis*, before ANOVA (Genstat, 2000).

## Results

### Laboratory experiments

*Laboratory Experiment 1: susceptibility of T. urticae to 40 isolates of entomopathogenic fungi.* The mean control mortalities of Replicates 1–3 were similar, being 22.8, 18.5 and 20.6%, respectively, at 6 days post inoculation (grand mean 20.4%, SD 10.41%). However, the mean control mortalities of the 12 blocks varied significantly ( $P < 0.01$ ), and were  $>25\%$  for four blocks,  $>30\%$  for two blocks, and  $>40\%$  for one block. Control mortalities were consistent within blocks with the exceptions of Block 1 of Replicate 2 and Block 3 of Replicate 3. Of the 40 fungal isolates examined, only three (*M. anisopliae* 442.99, *Hirsutella* sp. 457.99 and *V. lecanii* 450.99) caused more mortality ( $P < 0.05$ ) than the control (Table II). Two isolates (*H. thompsonii* 77.82 and *M. anisopliae* 445.99) apparently caused less mortality than the control. We have interpreted this as an artefact due to the variability of the data. We also analysed the data without Block 2 of Replicate 1, which was the block with the highest mean control mortality (41.7%), since it was felt that such a high control mortality could distort the results. However, doing this made no difference to the overall pathogenicity ranking of the isolates. The proportion of cadavers that supported sporulating mycelium was low for all isolates.

*Laboratory Experiment 2: susceptibility of T. urticae to six fungal isolates in a multiple dose bioassay.* The mean control mortalities of the nine blocks at 6 days post inoculation varied significantly ( $P < 0.05$ ) and ranged from 10.2 to 29.7% (grand mean 18.6%, SD 10.12%), but only exceeded 25% on one occasion. The generalised linear model (single line) used to describe the effect of conidia concentration on *T. urticae* mortality fitted the data well (Figure 1). *Beauveria bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, and *V. lecanii* 450.99 were all pathogenic to *T. urticae* (Table III). *Hirsutella* sp. 457.99 and *V. lecanii* 19.79 were not pathogenic to *T. urticae*, although previously, in Laboratory Experiment 1, *Hirsutella* sp. 457.99 caused more mortality than the control ( $P < 0.05$ ) (Table II). The non-pathogenicity of *Hirsutella*

Table II. Laboratory Experiment 1: pathogenicity of 40 isolates of entomopathogenic fungi applied to *T. urticae* at  $10^7$  mL<sup>-1</sup>, 6 days post inoculation. Isolates are ranked by estimated percentage mortality. Numbers in parenthesis represent the transformed data (logit transformation)

| Rank | Species                          | Isolate | Estimated % mortality | % Sporulating cadavers |
|------|----------------------------------|---------|-----------------------|------------------------|
| 1    | <i>Metarhizium anisopliae</i>    | 442.99  | 43.2 (−0.27)          | 9.7                    |
| 2    | <i>Hirsutella</i> sp.            | 457.99  | 38.0 (−0.49)          | 0.0                    |
| 3    | <i>Verticillium lecanii</i>      | 450.99  | 35.7 (−0.59)          | 22.2                   |
| 4    | <i>Beauveria bassiana</i>        | 455.99  | 33.0 (−0.71)          | 27.0                   |
| 5    | <i>Verticillium lecanii</i>      | 19.79   | 31.6 (−0.77)          | 28.6                   |
| 6    | <i>Hirsutella</i> sp.            | 440.99  | 30.7 (−0.82)          | 0.0                    |
| 7    | <i>Hirsutella thompsonii</i>     | 73.82   | 29.7 (−0.86)          | 6.7                    |
| 8    | <i>Hirsutella</i> sp.            | 436.99  | 29.1 (−0.89)          | 0.0                    |
| 9    | <i>Verticillium lecanii</i>      | 17.76   | 28.7 (−0.91)          | 26.0                   |
| 10   | <i>Beauveria bassiana</i>        | 431.99  | 27.2 (−0.99)          | 4.9                    |
| 11   | <i>Hirsutella thompsonii</i>     | 51.81   | 26.5 (−1.02)          | 0.0                    |
| 12   | <i>Hirsutella thompsonii</i>     | 75.82   | 25.9 (−1.05)          | 0.0                    |
| 13   | <i>Metarhizium anisopliae</i>    | 456.99  | 25.8 (−1.06)          | 12.9                   |
| 14   | <i>Beauveria bassiana</i>        | 433.99  | 25.7 (−1.06)          | 21.2                   |
| 15   | <i>Paecilomyces farinosus</i>    | 446.99  | 25.1 (−1.09)          | 1.0                    |
| 16   | <i>Verticillium lecanii</i>      | 453.99  | 24.7 (−1.11)          | 7.1                    |
| 17   | <i>Beauveria bassiana</i>        | 434.99  | 24.6 (−1.12)          | 29.4                   |
| 18   | <i>Metarhizium anisopliae</i>    | 441.99  | 24.6 (−1.12)          | 27.8                   |
| 19   | <i>Hirsutella necatrix</i>       | 49.81   | 24.5 (−1.13)          | 1.2                    |
| 20   | <i>Metarhizium anisopliae</i>    | 444.99  | 24.2 (−1.14)          | 11.6                   |
| 21   | <i>Metarhizium anisopliae</i>    | 392.93  | 24.2 (−1.15)          | 23.2                   |
| 22   | <i>Tolypocladium niveum</i>      | 449.99  | 23.6 (−1.17)          | 22.3                   |
| 23   | <i>Verticillium lecanii</i>      | 452.99  | 23.4 (−1.19)          | 11.0                   |
| 24   | <i>Beauveria bassiana</i>        | 454.99  | 23.0 (−1.21)          | 23.0                   |
| 25   | <i>Hirsutella</i> sp.            | 437.99  | 23.0 (−1.21)          | 0.0                    |
| 26   | Control                          | —       | 22.9 (−1.22)          | 0.00                   |
| 27   | <i>Paecilomyces fumosoroseus</i> | 409.96  | 22.3 (−1.25)          | 8.1                    |
| 28   | <i>Hirsutella thompsonii</i>     | 34.79   | 21.5 (−1.29)          | 2.1                    |
| 29   | <i>Hirsutella thompsonii</i>     | 71.82   | 21.3 (−1.31)          | 0.0                    |
| 30   | <i>Paecilomyces fumosoroseus</i> | 447.99  | 21.2 (−1.31)          | 8.8                    |
| 31   | <i>Hirsutella thompsonii</i>     | 463.99  | 20.8 (−1.34)          | 6.3                    |
| 32   | <i>Verticillium lecanii</i>      | 31.79   | 20.1 (−1.38)          | 21.7                   |
| 33   | <i>Beauveria bassiana</i>        | 432.99  | 19.3 (−1.43)          | 18.4                   |
| 34   | <i>Verticillium lecanii</i>      | 30.79   | 18.4 (−1.49)          | 12.1                   |
| 35   | <i>Verticillium lecanii</i>      | 1.72    | 16.9 (−1.60)          | 5.1                    |
| 36   | <i>Hirsutella</i> sp.            | 438.99  | 16.7 (−1.61)          | 0.0                    |
| 37   | <i>Beauveria bassiana</i>        | 460.99  | 15.5 (−1.69)          | 9.7                    |
| 38   | <i>Metarhizium anisopliae</i>    | 275.86  | 15.2 (−1.72)          | 8.0                    |
| 39   | <i>Tolypocladium inflatum</i>    | 448.99  | 14.6 (−1.77)          | 5.4                    |
| 40   | <i>Hirsutella thompsonii</i>     | 77.82   | 10.3 (−2.17)          | 0.0                    |
| 41   | <i>Metarhizium anisopliae</i>    | 445.99  | 6.7 (−2.64)           | 3.8                    |

Least significant difference between each isolate and control for estimated % mortality (transformed data), one-tailed test ( $P < 0.05$ ;  $df = 112$ ) = 0.590.

sp. 457.99 may be linked to the low germination of this isolate (average germination on SDA = 37%). For isolates *B. bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, *V. lecanii* 19.79 and *V. lecanii* 450.99, there was a linear relationship between the concentration of the conidia suspension and the dose of conidia presented onto the cover slips, estimated from cfu counts (data not shown). The  $R^2$  values were 0.988,

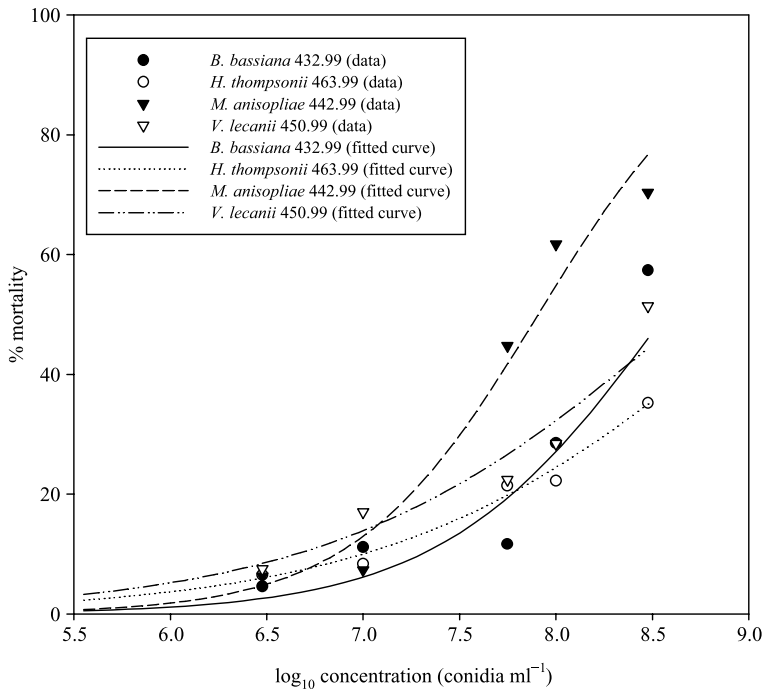


Figure 1. Relationship between percentage mortality at 6 days post inoculation and conidia concentration for adult female *T. urticae* treated with entomopathogenic fungi in Laboratory Experiment 2 (multiple-dose bioassay). Curves are derived from GLMs fitted to replicate data points for those isolates with fitted lines different from zero ( $P < 0.05$ ). Data points are the mean of three replicate bioassays.

0.990, 0.997, 0.996 and 0.996, respectively. Conidia of *Hirsutella* sp. 457.99 washed from cover slips produced no colonies on SDA, which is in keeping with the low germination observed for this isolate.

*Laboratory Experiment 3: comparison of direct application and indirect application bioassays.* Average control mortalities at 6 days post inoculation for the direct and indirect application methods were 32.8 and 25.1%, respectively (Table IV), and there was no difference between control mortalities for the two methods ( $P = 0.05$ ). There was no difference in percentage mortality between fungal isolates when the direct method was used ( $P = 0.05$ ). In contrast, *Naturalis-L* and *B. bassiana* 432.99 caused more mortality ( $P < 0.05$ ) than the other isolates when the indirect application method was used. The indirect application method increased ( $P < 0.05$ ) the percentage mortality of *T. urticae* treated with *B. bassiana* 432.99 and *Naturalis-L* compared with the direct application method. However, application method had no effect ( $P = 0.05$ ) on the percentage of *T. urticae* cadavers that supported sporulating mycelium, although there were differences between fungal isolates ( $P < 0.05$ ). Mean, angular transformed percentage sporulating cadavers for the isolates were: *B. bassiana* 432.99 = 29.2%, *Naturalis-L* = 24.2%, *H. thompsonii* 463.99 = 10.9%, *M. anisopliae* 442.99 = 35.4%, and *V. lecanii* 450.99 = 23.9% (41 degrees of freedom, least significant difference = 7.66).

Table III. Laboratory Experiment 2: estimated lethal concentrations (LC<sub>25</sub>) of six fungal isolates applied to adult female *T. urticae*. LC<sub>25</sub> values were calculated from mortality data recorded at 6 days post inoculation

| Species                       | Isolate | LC <sub>25</sub><br>(conidia mL <sup>-1</sup> ) | Fiducial limits (conidia mL <sup>-1</sup> ) |                    | Residual mean<br>deviance | Deviance<br>ratio | Slope of<br>regression line | SE of slope |
|-------------------------------|---------|-------------------------------------------------|---------------------------------------------|--------------------|---------------------------|-------------------|-----------------------------|-------------|
|                               |         |                                                 | Lower                                       | Upper              |                           |                   |                             |             |
| <i>Beauveria bassiana</i>     | 432.99  | $8.65 \times 10^7$                              | $1.47 \times 10^7$                          | $3.48 \times 10^8$ | 4.98                      | 8.79              | 1.74                        | 0.712       |
| <i>Hirsutella</i> sp.         | 457.99  | *                                               | *                                           | *                  | 4.77                      | 0.04              | *                           | *           |
| <i>Hirsutella thompsonii</i>  | 463.99  | $1.06 \times 10^8$                              | $2.87 \times 10^7$                          | $1.08 \times 10^9$ | 2.96                      | 7.93              | 1.07                        | 0.416       |
| <i>Metarhizium anisopliae</i> | 442.99  | $2.43 \times 10^7$                              | $1.21 \times 10^7$                          | $3.84 \times 10^7$ | 1.94                      | 53.18             | 2.10                        | 0.357       |
| <i>Verticillium lecanii</i>   | 19.79   | *                                               | *                                           | *                  | 3.45                      | 2.41              | *                           | *           |
| <i>Verticillium lecanii</i>   | 450.99  | $4.67 \times 10^7$                              | $1.12 \times 10^7$                          | $1.28 \times 10^8$ | 2.46                      | 10.95             | 1.08                        | 0.356       |

Residual degrees of freedom = 12 for *Hirsutella* sp. 457.99, and 13 for all other isolates.

\*The LC<sub>25</sub> could not be estimated because the slope of the fitted line was not significantly different from zero ( $P < 0.05$ ).

Table IV. Laboratory Experiment 3: mean percentage mortality at 6 days post inoculation of adult female *T. urticae* treated with entomopathogenic fungi using direct and indirect application methods. Conidia were applied at a concentration of  $1 \times 10^8 \text{ mL}^{-1}$ . Numbers in parenthesis represent the transformed data (angular transformation). Least significant differences (LSDs) correspond to the transformed data

| Species                       | Isolate     | Direct spray | Indirect spray | LSD <sup>†</sup> |
|-------------------------------|-------------|--------------|----------------|------------------|
| Control                       | –           | 32.8 (34.4)  | 25.1 (29.8)    | (9.37)           |
| <i>Beauveria bassiana</i>     | 432.99      | 46.2 (42.8)  | 72.2 (63.7)    | (13.25)          |
| <i>Beauveria bassiana</i>     | Naturalis-L | 52.1 (46.7)  | 95.2 (79.2)    | (13.25)          |
| <i>Hirsutella thompsonii</i>  | 463.99      | 37.6 (37.7)  | 54.9 (47.9)    | (13.25)          |
| <i>Metarhizium anisopliae</i> | 442.99      | 54.4 (47.6)  | 48.0 (43.4)    | (13.25)          |
| <i>Verticillium lecanii</i>   | 450.99      | 51.8 (46.1)  | 55.7 (48.4)    | (13.25)          |
| LSD <sup>††</sup>             |             | (11.48)      | (11.48)        |                  |

<sup>†</sup>Least significant difference between direct spray and indirect spray ( $P < 0.05$ ;  $df = 62$ ).

<sup>††</sup>Least significant difference between isolates ( $P < 0.05$ ;  $df = 62$ ).

### Glasshouse experiments

*Glasshouse Experiment 1: efficacy of entomopathogenic fungi against T. urticae populations on tomato.* There was a trend for increasing numbers of motile *T. urticae* higher up the plants (Table V). At the top sample position, four of the treatments (*H. thompsonii* 463.99, *M. anisopliae* 442.99, *V. lecanii* 450.99, and Naturalis-L) reduced ( $P < 0.05$ ) the numbers of motile *T. urticae* and eggs per leaf compared with the untreated control. Naturalis-L reduced the numbers of motile *T. urticae* by 97% at the top sample position. At the middle sample position, all the treatments reduced ( $P < 0.05$ ) the numbers of motile *T. urticae* compared to the control, and *V. lecanii* 450.99 and Naturalis-L reduced ( $P < 0.05$ ) the numbers of *T. urticae* eggs. Finally, at the bottom sample position, all the treatments reduced ( $P < 0.05$ ) the numbers of motile *T. urticae* compared with the control, and four treatments (*B. bassiana* 432.99, *H. thompsonii* 463.99, *M. anisopliae* 442.99, and Naturalis-L) reduced ( $P < 0.05$ ) the numbers of *T. urticae* eggs. Overall, Naturalis-L had the greatest effect on the *T. urticae* population.

*Glasshouse Experiment 2: efficacy of Naturalis-L as a supplementary treatment to the predatory mite, P. persimilis, for control of T. urticae populations on tomato.* Application of *P. persimilis* on its own did not reduce numbers of *T. urticae* adults, nymphs or eggs ( $P = 0.05$ ) (Table VI). Application of *P. persimilis* + fenbutatin oxide reduced ( $P < 0.05$ ) the numbers of *T. urticae* nymphs (80% reduction), but did not reduce ( $P = 0.05$ ) the numbers of adults or eggs compared with the untreated control or the *P. persimilis* treatment. In contrast, application of *P. persimilis* + Naturalis-L reduced ( $P < 0.05$ ) numbers of *T. urticae* adults, nymphs and eggs compared with all other treatments (98% reduction in all three cases). Fewer ( $P < 0.05$ ) *P. persimilis* were recorded from both the *P. persimilis* + fenbutatin oxide treatment, and the *P. persimilis* + Naturalis-L treatment, than the *P. persimilis* – only treatment.

### Discussion

*Tetranychus urticae* is a serious pest of crops because of its capacity to develop resistance to chemical pesticides, but little work has been done on microbial control as an alternative to chemical treatments (Chandler et al., 2000). This is the first study in

Table V. Glasshouse Experiment 1: mean (back transformed) total number of motile (adults+nymphs) *T. urticae* and eggs found per leaf on tomato plants after treatment with entomopathogenic fungi. Numbers in parenthesis represent the transformed data ( $\log_{10}(\text{total}+0.375)$  transformation). Least significant differences (LSDs) correspond to the transformed data

| Treatment                                         | Top of plant |             | Middle of plant |             | Bottom of plant |             |
|---------------------------------------------------|--------------|-------------|-----------------|-------------|-----------------|-------------|
|                                                   | Motile       | Eggs        | Motile          | Eggs        | Motile          | Eggs        |
| Control                                           | 83.0 (1.92)  | 10.2 (1.02) | 6.7 (0.85)      | 1.8 (0.34)  | 4.0 (0.64)      | 1.1 (0.16)  |
| <i>B. bassiana</i> 432.99                         | 24.2 (1.39)  | 4.3 (0.67)  | 1.7 (0.31)      | 1.0 (0.14)  | 0.1 (−0.30)     | 0.2 (−0.28) |
| <i>B. bassiana</i> Naturalis-L                    | 2.4 (0.44)   | 2.8 (0.50)  | 0.5 (−0.06)     | 0.7 (0.03)  | 0.6 (−0.01)     | 0.3 (−0.18) |
| <i>H. thompsonii</i> 463.99                       | 4.5 (0.69)   | 2.1 (0.40)  | 2.3 (0.43)      | 0.9 (0.12)  | 1.2 (0.21)      | 0.3 (−0.16) |
| <i>M. anisopliae</i> 442.99                       | 11.8 (1.09)  | 2.5 (0.46)  | 2.3 (0.43)      | 1.0 (0.14)  | 1.1 (0.16)      | 0.7 (0.02)  |
| <i>V. lecanii</i> 450.99                          | 9.2 (0.98)   | 1.4 (0.24)  | 1.3 (0.23)      | 0.5 (−0.08) | 0.8 (0.06)      | 0.3 (−0.18) |
| LSD between treatments ( $P < 0.05$ ; $df = 15$ ) | (0.591)      | (0.495)     | (0.393)         | (0.296)     | (0.225)         | (0.199)     |

Table VI. Glasshouse Experiment 2: mean (backtransformed) number of *T. urticae* adults, nymphs and eggs per leaf following treatment with *Phytoseiulus persimilis* with or without fenbutatin oxide or Naturalis-L. Numbers in parenthesis represent the transformed data ( $\log_{10}(\text{total} + 0.375)$  transformation). Least significant differences (LSDs) between treatments correspond to the transformed data

|                                                                       | Adults      | Nymphs       | Eggs         | <i>Phytoseiulus persimilis</i> |
|-----------------------------------------------------------------------|-------------|--------------|--------------|--------------------------------|
| Control                                                               | 38.2 (1.59) | 484.9 (2.69) | 537.9 (2.26) | –                              |
| <i>P. persimilis</i>                                                  | 39.9 (1.60) | 210.5 (2.32) | 96.7 (1.99)  | 2.9 (0.52)                     |
| <i>P. persimilis</i> + fenbutatin oxide                               | 40.5 (1.61) | 101.5 (2.01) | 115.5 (2.06) | 0.8 (0.06)                     |
| <i>P. persimilis</i> + Naturalis-L                                    | 0.8 (0.69)  | 11.5 (1.08)  | 9.1 (0.98)   | 0.5 (–0.06)                    |
| LSD between treatments ( $P < 0.05$ )                                 | (0.656)     | (0.5288)     | (0.756)      | (0.440)                        |
| df = 8 for <i>P. persimilis</i> treatment,<br>11 for other treatments |             |              |              |                                |

which a wide range of species and isolates of anamorphic entomopathogenic fungi have been compared against tetranychid mites, although isolates of *B. bassiana*, *H. thompsonii*, and *V. lecanii* have been reported previously as highly pathogenic (Gerson et al., 1979; Gardner et al., 1982; Gillespie et al., 1982; Wright & Kennedy, 1996; Alves et al., 2002; Irigaray et al., 2003). However, in our study, only low levels of mortality were observed with these and other fungi in laboratory bioassays. This may have been due to the action of tomato allelochemicals which can inhibit entomopathogenic fungi, although effects vary with fungal isolate, host species and tomato variety (Costa & Gaugler, 1989; Vidal et al., 1998; Poprawski et al., 2000). There was also evidence that a proportion of mites killed were not colonised by fungi, suggesting that limited growth or toxicosis could be sufficient to cause mortality (Butt et al., 1992). The variability in the results of the laboratory bioassays made isolate selection problematic, gave some misleading results, and led us to adopt a control-heavy design early in the programme. Even with this, variation in control mortalities between blocks was still an issue. High control mortalities were a particular concern because they might be associated with an interaction between fungus and non-fungus mortality factors in a treatment bioassay. However, low control mortalities may be difficult to achieve with tetranychid mites, which have short natural life spans and may be very responsive to stress factors under bioassay conditions. For example, the average longevity of adult female *T. urticae* from life history studies is only 13–15 days at 24–25°C, and survivorship can be reduced by the type of host plant (Carey & Bradley, 1982; So & Takafuji, 1991; Belczewski & Harmsen, 2002). Unfortunately, it is not possible to compare our control mortality data with the other reports of fungal bioassays against tetranychid mites, as bioassays have either not been replicated (Gerson et al., 1979) or control mortalities have not been presented (Alves et al., 2002; Irigaray et al., 2003).

We found that the method used to apply fungal conidia in the laboratory bioassay significantly affected the efficacy of Naturalis-L and *B. bassiana* 432.99 (which was cultured from Naturalis-L), although it had no effect on *H. thompsonii* 463.99, *M. anisopliae* 442.99, and *V. lecanii* 19.79. The reason why application method only affected the efficacy of *B. bassiana* 432.99/Naturalis-L is not known. The ability of conidia to attach to the host cuticle is strongly correlated with virulence (Altre et al., 1999), and so it is possible that more conidia of this fungus attached to *T. urticae* with the indirect application technique. However, application method can also influence the pattern of deposition of conidia on the cuticle, and this can have a greater effect on

fungus efficacy than the total number of conidia deposited. For example, Fernandez et al. (2001) found that when larvae of the Colorado potato beetle, *Leptinotarsa decemlineata*, were exposed to *B. bassiana*-treated foliage, they acquired more conidia, but showed lower mortality, than larvae sprayed with a conidia suspension. This was because conidia acquired from the foliage attached predominantly to the ventral surface of the larvae, which was a poor site for infection. We chose a direct application bioassay at the start of this study because it enables a precise dose of conidia to be presented, which is thought to be important for reducing variability and enabling results to be repeated (Goettel & Inglis, 1997; Butt & Goettel, 2000). However, the use of a direct application bioassay in this case could have led to mistakes in isolate selection and the indirect bioassay would appear to be preferable.

The fungal isolates examined in the glasshouse studies gave much better control of *T. urticae* than expected from the laboratory bioassays. It is noteworthy that *M. anisopliae* isolate 442.99 was effective in the glasshouse and was also the most virulent isolate in the direct application laboratory bioassays, as this species has not been reported before as a pathogen of tetranychid mites. However, with the exception of Naturalis-L, all the fungi were applied in the glasshouse at a high concentration ( $1 \times 10^8 \text{ mL}^{-1}$ ), which is unlikely to be economic in a commercial product. The best glasshouse results were observed with Naturalis-L, which reduced *T. urticae* populations by up to 97% on its own, and caused up to 98% reduction within 7 days when applied with *P. persimilis*. It also out-performed fenbutatin oxide as a supplementary treatment to *P. persimilis*. Elsewhere, Naturalis has been reported to reduce populations of *T. urticae* on roses (Wright & Kennedy, 1996), cotton (Hinz & Wright, 1997) and goutweed (Abbey & Pundt, 1998). Naturalis-L is reported by its manufacturers to have no significant effects on a range of predators/parasitoids, including species of *Geocoris*, *Encarsia*, *Eretmocerus* and *Chrysoperla* (Wright & Knauf, 1994). In our glasshouse experiments, both Naturalis-L and fenbutatin oxide caused a reduction in the numbers of *P. persimilis*, but we do not know if this was caused directly by the products killing *P. persimilis* or whether the absence of prey caused them to migrate. A key finding in our study was that Naturalis-L had a markedly greater efficacy than the laboratory isolate, *B. bassiana* 432.99, cultured from it. We assume that the difference in efficacy was due to the commercial formulation used for Naturalis-L. Formulation is known to increase the efficacy of microbial biopesticides by improving application and coverage, activity, and persistence on the leaf surface (Jones et al., 1997), and hence it is possible that the efficacy of the other isolates could have been improved significantly using the right formulation. Unfortunately, in this study, we were not able to examine the Naturalis-L formulation without the fungus, to see if it had any effect on spider mites, which could have helped explain the high efficacy of the product. We examined other isolates cultured from commercial products in the laboratory bioassays, as opposed to using the commercial products directly, in order to compare their intrinsic virulences, but this may have caused us to overlook potentially useful commercial products. Only two products are currently available to growers in the UK, Mycotal and Vertalec (Koppert BV, The Netherlands), although Naturalis-L is going through UK registration.

Our findings suggest that Naturalis-L has potential as a control agent of *T. urticae* on tomato crops, although further work is required on its mode of action and its compatibility with *P. persimilis*. Even if Naturalis-L has some activity against *P. persimilis*, this may not prevent its use as a supplementary treatment provided



that it does not persist in the crop, and *P. persimilis* can be introduced quickly once the fungus has reduced *T. urticae* populations to below the economic threshold. If the compatibility of Naturalis-L with other natural enemies turns out to be a problem, there are opportunities to investigate other fungal biopesticides against *T. urticae*, which may have narrower host ranges. Future investigations are also required to explain why fungal isolates gave better control of *T. urticae* in the glasshouse than expected from the laboratory bioassays. This should lead to improvements in the bioassay and make the selection of virulent isolates more reliable.

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